

SOC/L1 - 7.2. Network Discovery Detection

Network Discovery Detection

Task 1 - Introduction

Whenever an attacker wants to target a network, they first have to better understand their target. To do this, attackers perform certain actions to discover the target network. This is often one of the first steps of detecting an attacker's activity and perhaps one of the most common stages of attack a SOC analyst observes in their shift.

Learning Objectives:

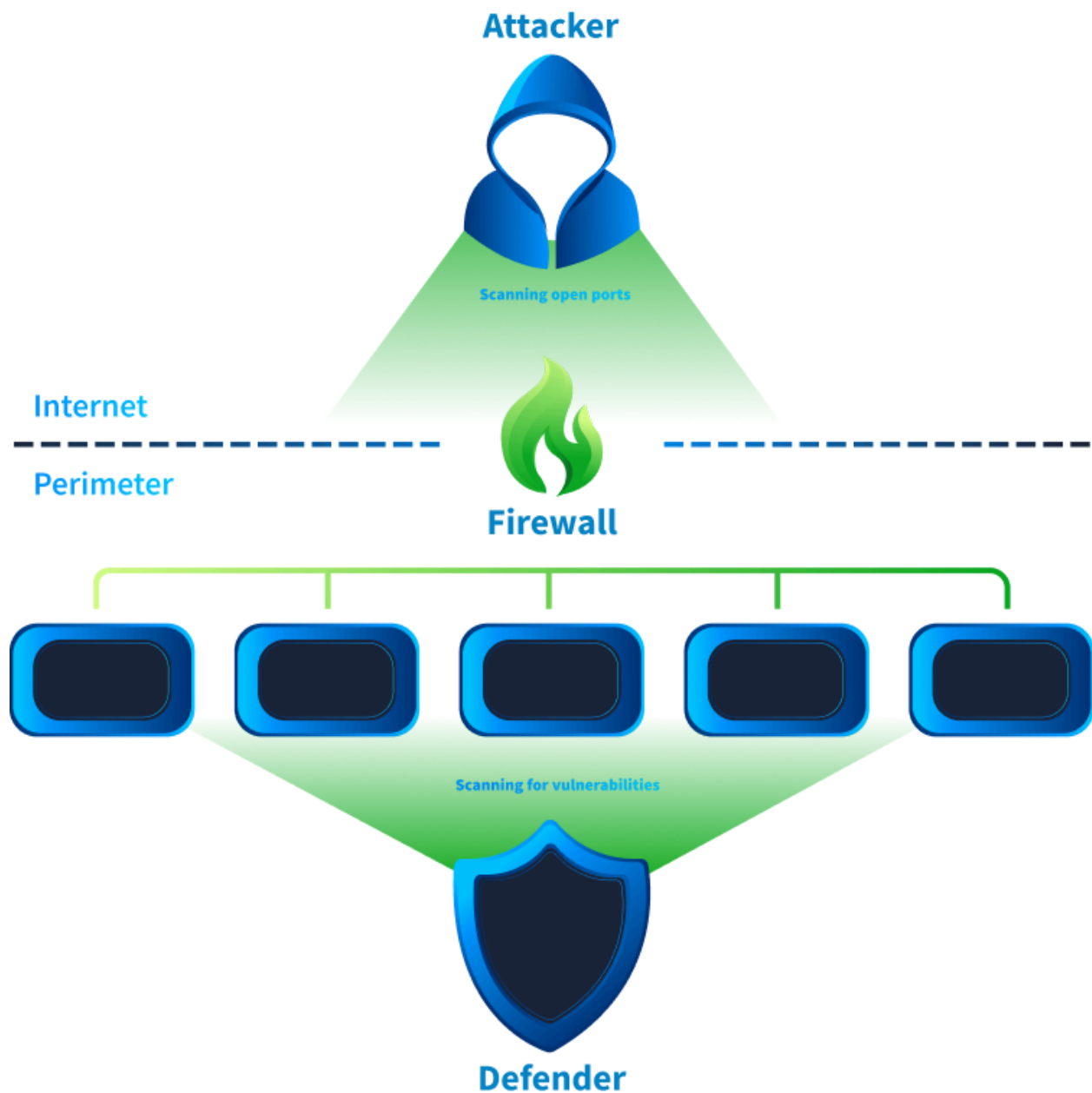
- What is network discovery
- Why attackers perform network discovery
- What are the different types of network discovery
- How network discovery techniques work, and how we can detect them

Task 2 - Network Discovery

Attackers and Network Discovery: Attackers start their attack by trying to discover an organisation's assets that are open to the publicly accessible internet (called the attack surface). During this discovery phase, the attacker attempts to ascertain some of the following information:

- What assets can be accessed by the attacker?
- What are the IP addresses, ports, OS, and services running on these assets?
- What versions of services are running? Does any service have a vulnerability that can be exploited?

In short, the attacker is trying to find an opening that will allow them to exploit the network.



Defenders and Network Discovery: Defenders also sometimes run software that performs network discovery activity. During this activity, defenders want to achieve the following goals:

- Inventory the organisation's assets and ensure all assets are documented.
- Ensure no unnecessary IP, port, or service is open, and whatever is running is necessary.

- Ensure vulnerabilities are patched, or at least the exploitable vulnerabilities are patched.

In short, defenders attempt to reduce the attack surface as much as possible.

The Challenge in Detecting Network Discovery

- Allowlist known internal and benign external scanners, ensuring no alerts are triggered on those sources.
- Integrate Threat Intelligence with detection use cases and flag scanning activity only from known malicious or suspicious sources.
- Since the previous point has a chance of missing some malicious scanning activity, some teams use the Threat Intelligence to raise the severity of the alerts instead of only triggering alerts on them.

Answer the questions below

What do attackers scan, other than, IP addresses, ports, and OS version, in order to identify vulnerabilities in a network?

Services

Task 3 - External vs Internal Scanning

External Scanning Activity: An attacker scans a company's public-facing perimeter from an external IP to find entry points. Because the attacker lacks a network foothold, this initial reconnaissance is low-severity and can be mitigated by blocking the external IP on the perimeter firewall (though the attacker may return using a new IP).

Internal Scanning Activity: An internal-to-internal scan uses private IPs inside the network, signaling an attacker already has a foothold and is performing reconnaissance for lateral movement. Because the network is breached, this is a high-severity alert that requires escalating to Incident Response and performing a root cause analysis—not just blocking the IP.

Identifying Internal and External Scanning: Navigate to `/home/ubuntu/Downloads/logs`. There will be a few `csv` files in this directory. A `csv` (Comma-Separated Values) file is a plain-text format that stores tabular data by using lines for rows and commas to separate columns. The Linux `cut` command is a command-line utility

used to extract specific columns or fields out of structured log files or text streams.

Answer the questions below

Which file contains logs that showcase internal scanning activity?

log-session-2.csv

run the query: `head -n2 log-session-1.csv`

```
ubuntu@tryhackme:~/Downloads/logs$ head -n2 log-session-1.csv
"@timestamp","source.ip","source.port","destination.ip","destination.port","rule
.name","rule.category","rule.action","network.protocol",message,"event.dataset"
"Sep 7, 2025 @ 17:16:42.944","203.0.113.25",39120,"192.168.230.145",5922,"-","-
","-","-","{"ts":1757265402.944286,"uid":"CjxnoPYeUTSU7IAo","id.orig_h":
"203.0.113.25","id.orig_p":39120,"id.resp_h":"192.168.230.145","id.resp_
p":5922,"proto":"tcp","conn_state":"S0","local_orig":true,"local_res
p":true,"missed_bytes":0,"history":"S","orig_pkts":1,"orig_ip_bytes":
44,"resp_pkts":0,"resp_ip_bytes":0,"community_id":"1:JbLPvC3YPB/75Ez5qzk/
uYmWvg4=","orig_mac_oui":"VMware, Inc."},"zeek.conn"
```

How many log entries are present for the internal IP performing internal scanning activity?

2276

run the query: `cat log-session-2.csv |cut -d "," -f3 |uniq -c`

```
ubuntu@tryhackme:~/Downloads/logs$ cat log-session-2.csv |cut -d "," -f3 |uniq -
c
    1 "source.port"
  2276 "192.168.230.127"
```

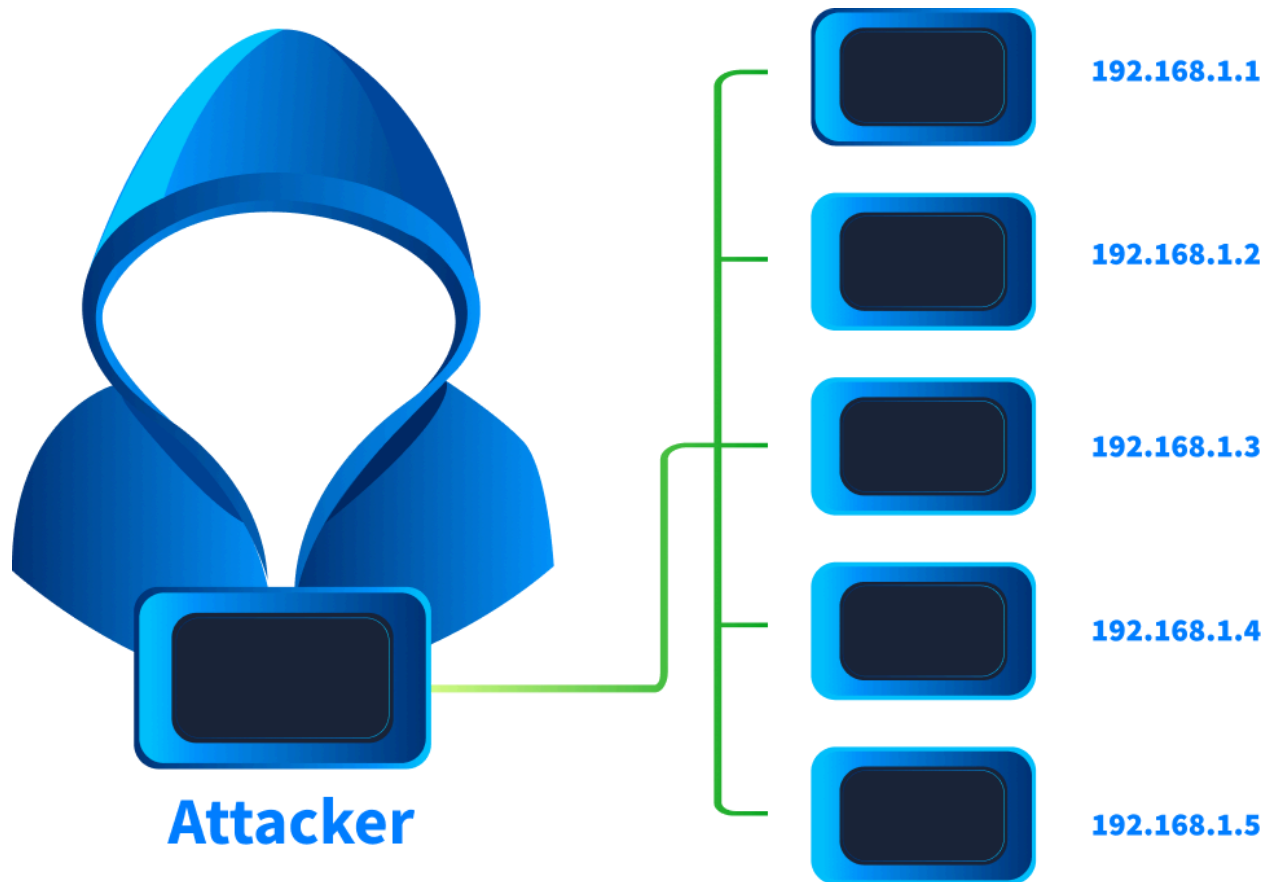
What is the external IP address that is performing external scanning activity?

203.0.113.25

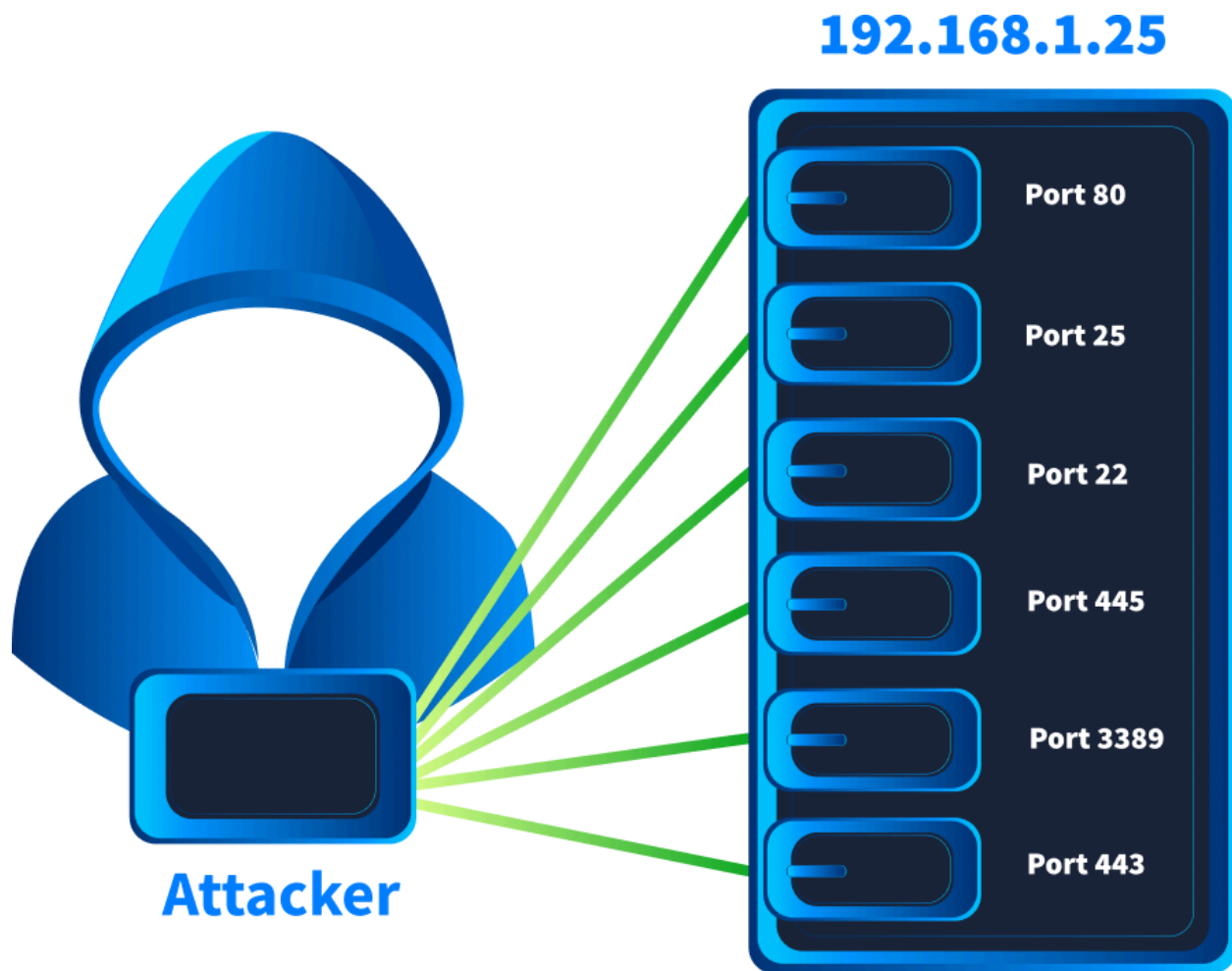
```
ubuntu@tryhackme:~/Downloads/logs$ cat log-session-0.csv |cut -d "," -f3 |uniq -
c
    1 "source.port"
  2014 "203.0.113.25"
```

Task 4 - Horizontal vs Vertical Scanning

Horizontal Scanning: Horizontal scanning occurs when an attacker targets a single port across multiple IP addresses to discover vulnerable systems exposing that specific service.



Vertical Scanning: Vertical scanning occurs when an attacker probes multiple ports on a single IP address to identify all open services and footprint a high-value target.



Answer the questions below

One of the log files contains evidence of a horizontal scan. Which IP range was scanned?

203.0.113.0/24

In the same log file, there is one IP address on which a vertical scan is performed. Which IP address is this?

192.168.230.145

On one of the IP addresses, only a few ports are scanned which host common services. Which are the ports that are scanned on this IP address?

80, 445, 3389 (Open the Excel file and analyze the questions above)

Task 5 - The Mechanics of Scanning

Ping Sweep: Ping sweeps are a basic network discovery technique used to identify active hosts on a network. They work by sending **ICMP packets** and checking for responses from online devices. However, many organisations block ICMP traffic, making ping sweeps easier to prevent or detect.

TCP SYN Scans: TCP SYN scans use the **TCP three-way handshake (SYN, SYN-ACK, ACK)** to identify active hosts and open ports. A SYN-ACK response indicates that the host is online and the port is open. This technique is considered stealthy because it can blend with normal network traffic, making detection more difficult.

UDP Scan: UDP scans identify open ports by sending UDP packets and analysing responses. An ICMP "port unreachable" message indicates a closed port, while no response may suggest an open port. However, UDP scans are slower and less reliable because they depend on waiting for responses or timeouts. SOC analysts should understand authorised internal scans, including their source IPs, scan types, and schedules, to reduce false positives and avoid unnecessary alerts.

Identifying Scan Types: Open a browser inside the attached VM and use the following credentials. Navigate to the hamburger menu on the top-left side, and click on Discover.

Answer the questions below

| Which source IP performs a ping sweep attack across a whole subnet?

192.168.230.127 (Filter ICMP traffic, as ping sweeps use that protocol. ICMP (Internet Control Message Protocol) is a network-layer protocol used by devices to communicate operational status,)

elastic Find apps, content, and more.

Discover Check out context-aware Discover Try ES|QL Inspect Alerts + Save

Data view All logs Filter your data using KQL syntax Sep 4, 2025 @ 02:52:00.3... → Sep 7, 2025 @ 17:16:42.944

network.protocol.keyword: icmp

Search field names 0

Selected fields 5: source.ip, source.port, destination.ip, destination.port, network.protocol

Popular fields 6: rule.name, destination.ip, destination.port

Documents (256) Field statistics

@timestamp	source.ip	source.port	destination.ip	destination.port	network.protocol
Sep 7, 2025 @ 17:10:07.804230000	192.168.230.127	8	192.168.230.145	-	icmp
Sep 7, 2025 @ 17:10:04.845801000	192.168.230.127	8	192.168.230.1	-	icmp
Sep 7, 2025 @ 17:09:11.220509000	192.168.230.127	8	203.0.113.254	-	icmp
Sep 7, 2025 @ 17:09:11.200951000	192.168.230.127	8	203.0.113.253	-	icmp

The zeek.conn.conn_state value shows the connection state. Using the information provided by this value, identify the type of scan being performed by 203.0.113.25 against 192.168.230.145

TCP SYN Scan (Filter network protocol)

Data view All logs Filter your data using KQL syntax Sep 4, 2025 @ 02:52:00.3... → Sep 7, 2025 @ 17:16:42.944

source.ip: 203.0.113.25

Search field names 0

Selected fields: zeek.conn.alert.metadata.updated_at, zeek.conn.alert.rev, zeek.conn.alert.rule, zeek.conn.alert.severity, zeek.conn.alert.signature, zeek.conn.alert.signature_id, zeek.conn.capture_file, zeek.conn.community_id, zeek.conn.conn_state, zeek.conn.dest_ip, zeek.conn.dest_port, zeek.conn.direction, zeek.conn.duration, zeek.conn.event_type, zeek.conn.flow_id, zeek.conn.history, zeek.conn.id.orig_h

Documents (2,004) Field statistics

@timestamp	source.ip	source.port	destination.ip	destination.port	network.protocol	zeek.conn.conn_state
Sep 7, 2025 @ 17:16:42.941845000	203.0.113.25	39,120	192.168.230.145	5,850	tcp	S0
Sep 7, 2025 @ 17:16:42.939535000	203.0.113.25	39,120	192.168.230.145	9,500	tcp	S0
Sep 7, 2025 @ 17:16:42.937283000	203.0.113.25	39,120	192.168.230.145	57,797	tcp	S0
Sep 7, 2025 @ 17:16:42.934918000	203.0.113.25	39,120	192.168.230.145	8,333	tcp	S0
Sep 7, 2025 @ 17:16:42.932667000	203.0.113.25	39,120	192.168.230.145	6,567	tcp	S0
Sep 7, 2025 @ 17:16:42.930297000	203.0.113.25	39,120	192.168.230.145	8,100	tcp	S0
Sep 7, 2025 @ 17:16:42.927790000	203.0.113.25	39,120	192.168.230.145	416	tcp	S0

Is there any UDP scanning attempt in the logs? N

Task 6 - Conclusion

- What is network discovery
- The difference between external and internal scanning, and the severity on each.
- Port scanning and host scanning, and why each of those is performed.
- How different types of scans are performed at a more granular room.